

SmolVLM for Dense Video Captioning

Week 12: Internship Update

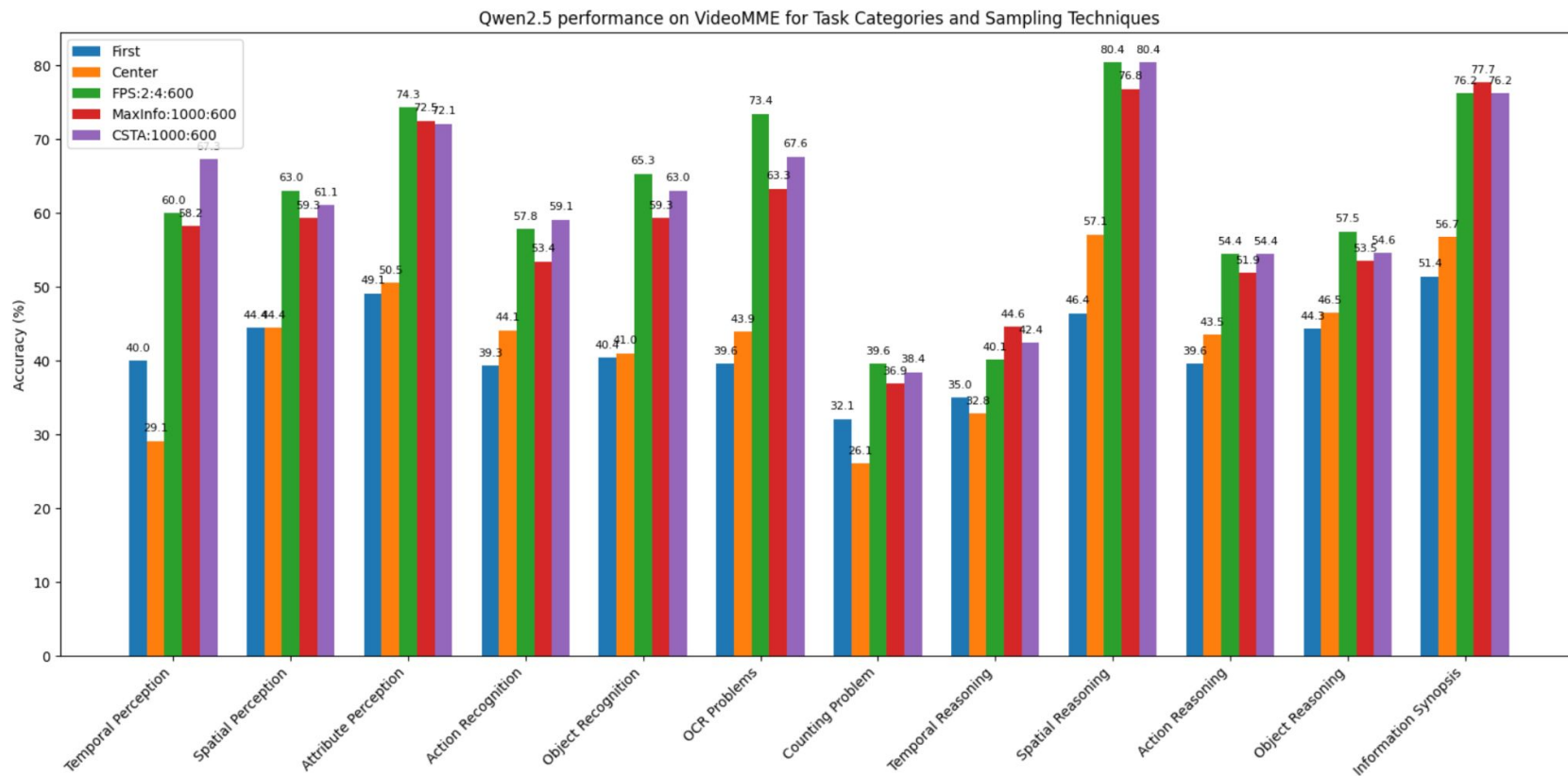
Qwen2.5-VL-3B-Instruct

- MaxInfo:1000:600, average number of frames:87.67, execution time:75062.48s
- CSTA:1000:600, average number of frames:148.92, execution time:80961.79s
- FPS:2:4:600, execution time:82485.69s

Video Type	First	Center	FPS:2:4:600	MaxInfo:1000:600	CSTA:1000:600
Short	46.0%	47.9%	72.6%	68.1%	70.2%
Medium	40.1%	40.0%	58.7%	54.9%	57.7%
Long	39.4%	42.6%	49.9%	49.6%	50.3%
Entire dataset	41.9%	43.5%	60.4%	57.5%	59.4%

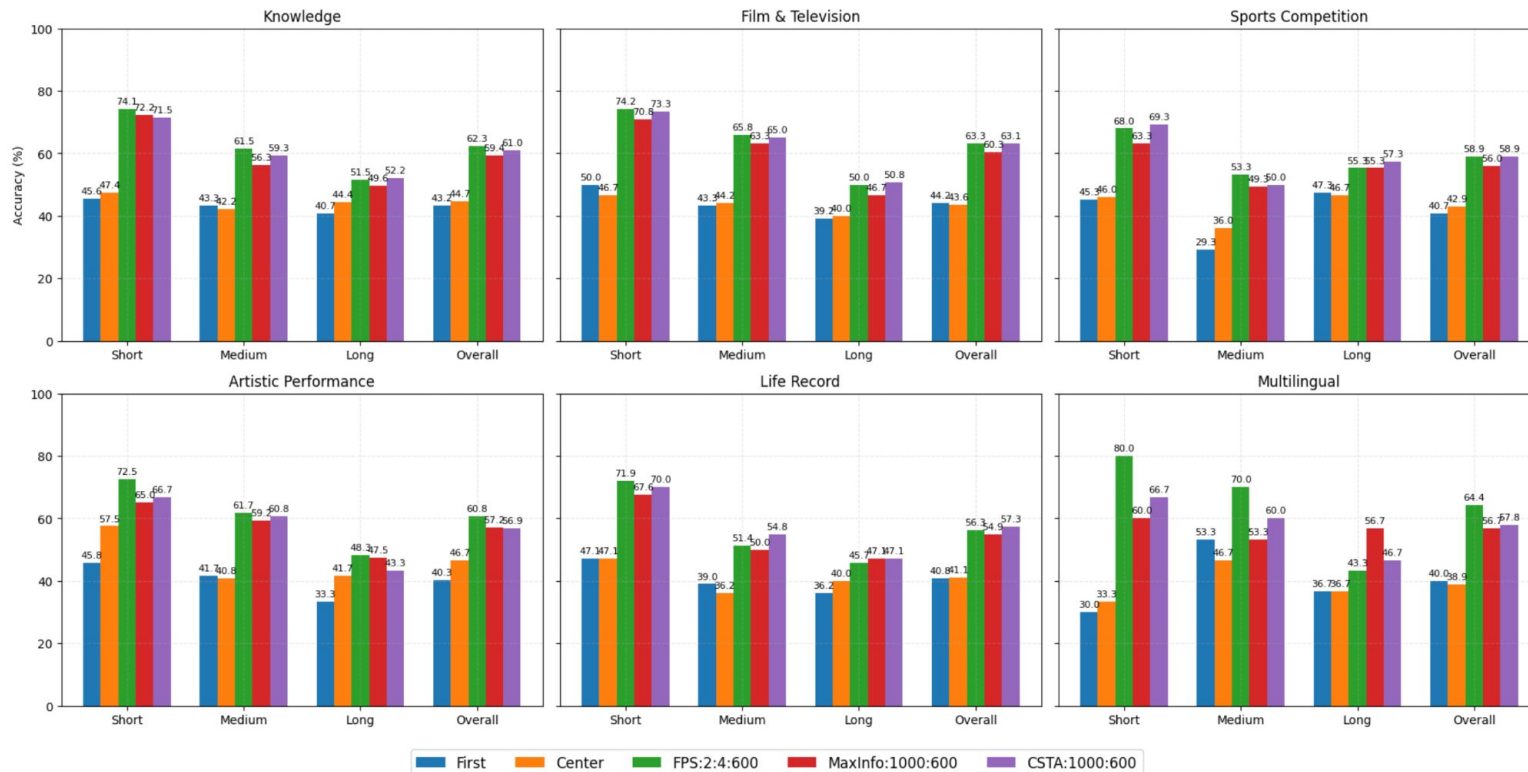
Table 1: VideoMME results over frame sampling techniques: Model: Qwen2.5

Qwen2.5-VL-3B-Instruct



Qwen2.5-VL-3B-Instruct

Qwen2.5 performance on VideoMME for Video Categories and Sampling Techniques



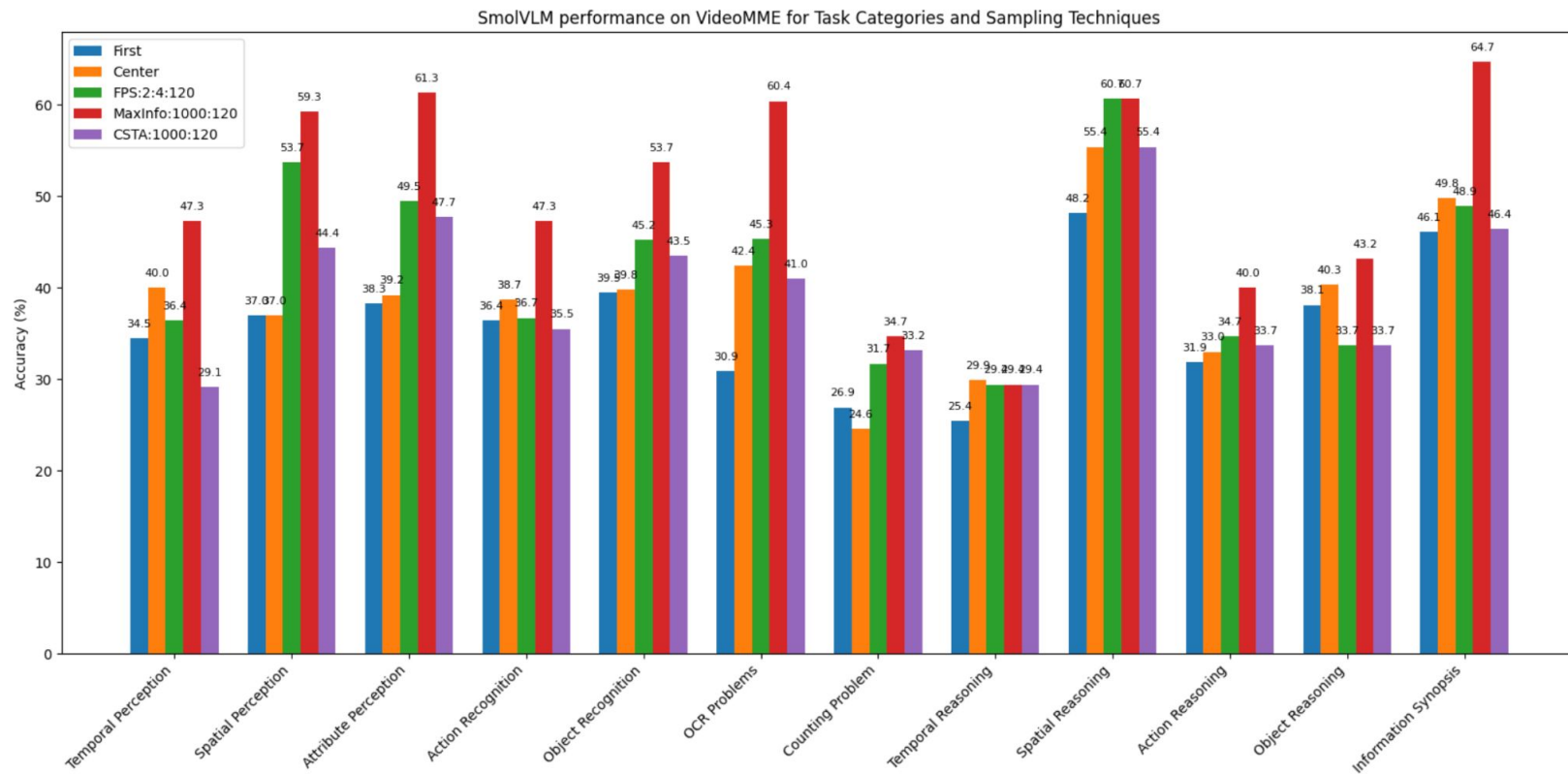
SmolVLM2-2.2B-Instruct

- MaxInfo:1000:120, average number of frames:87.67, execution time:76690.23s
- CSTA:1000:120, average number of frames:119.49, execution time:79160.45s
- FPS:2:4:120, execution time:22493.18s

Video Type	First	Center	FPS:2:4:120	MaxInfo:1000:120	CSTA:1000:120
Short	38.4%	41.4%	50.1%	58.6%	48.6%
Medium	37.4%	40.4%	37.2%	48.6%	35.6%
Long	32.8%	33.4%	32.4%	38.9%	31.3%
Entire dataset	36.2%	38.4%	39.9%	48.7%	38.5%

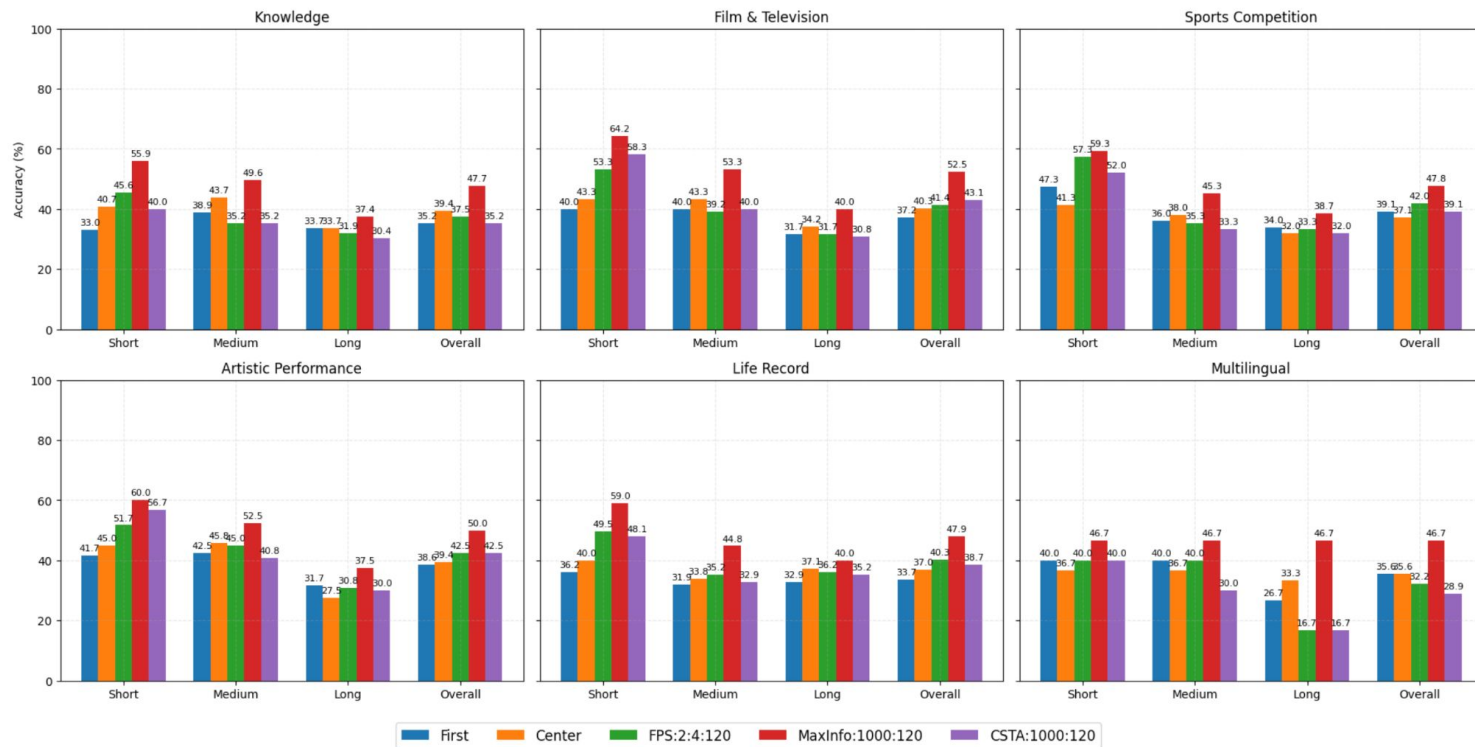
Table 3: VideoMME results over frame sampling techniques: Model: SmolVLM

SmolVLM2-2.2B-Instruct



SmolVLM2-2.2B-Instruct

SmolVLM performance on VideoMME for Video Categories and Sampling Techniques



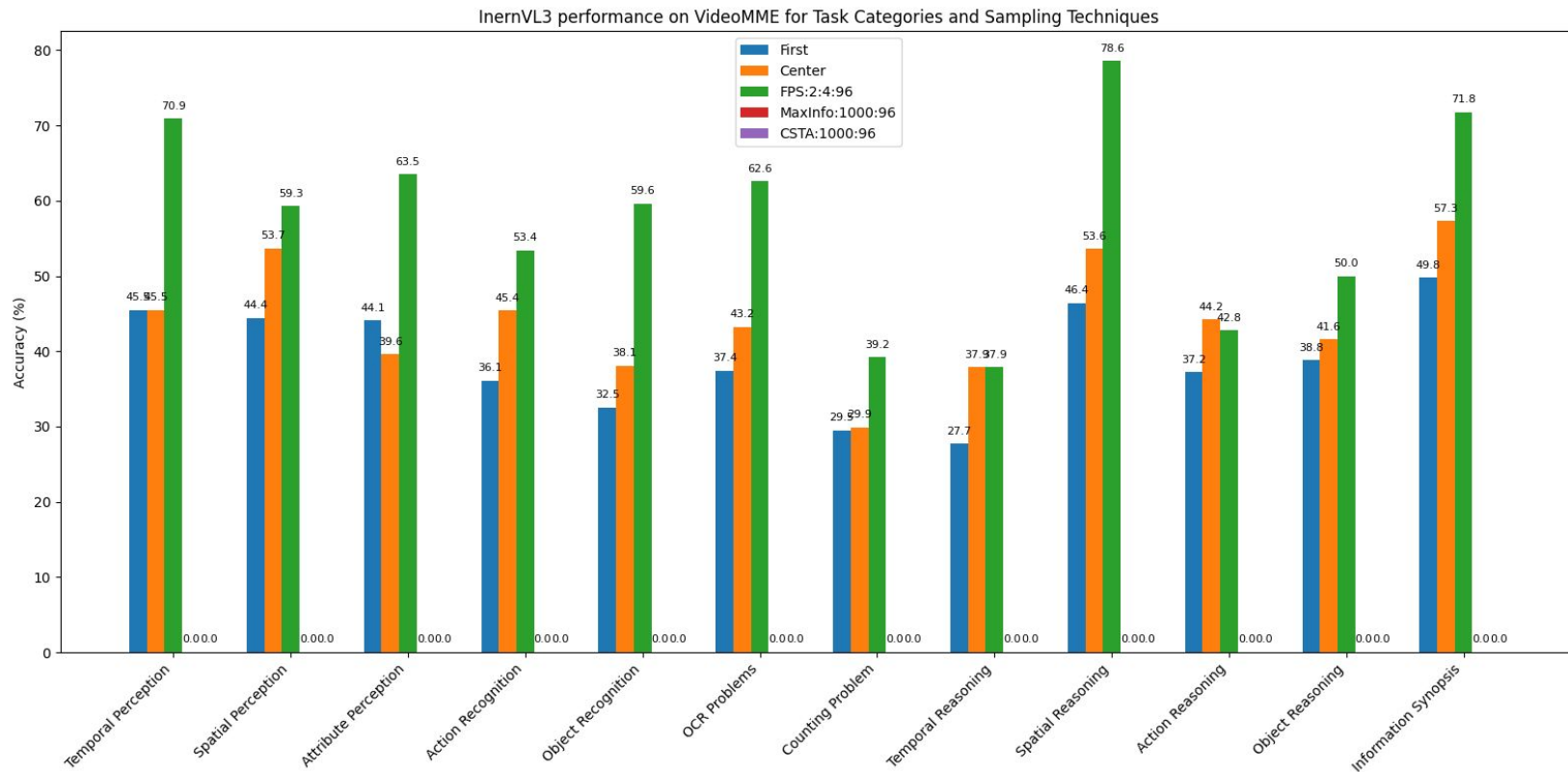
InternVL3-2B

- MaxInfo:1000:96
- CSTA:1000:96
- FPS:2:4:96, execution time:22493.18s, FPS:2:4:32, execution time:7886.20s

Video Type	First	Center	FPS:2:4:96	MaxInfo:1000:96	CSTA:1000:96
Short	41.3%	47.3%	67.3(67.4)%	%	%
Medium	38.2%	41.7%	53.7(51.1)%	%	%
Long	34.2%	39.4%	42.8(42.4)%	%	%
Entire dataset	37.9%	42.8%	54.6(53.7)%	%	%

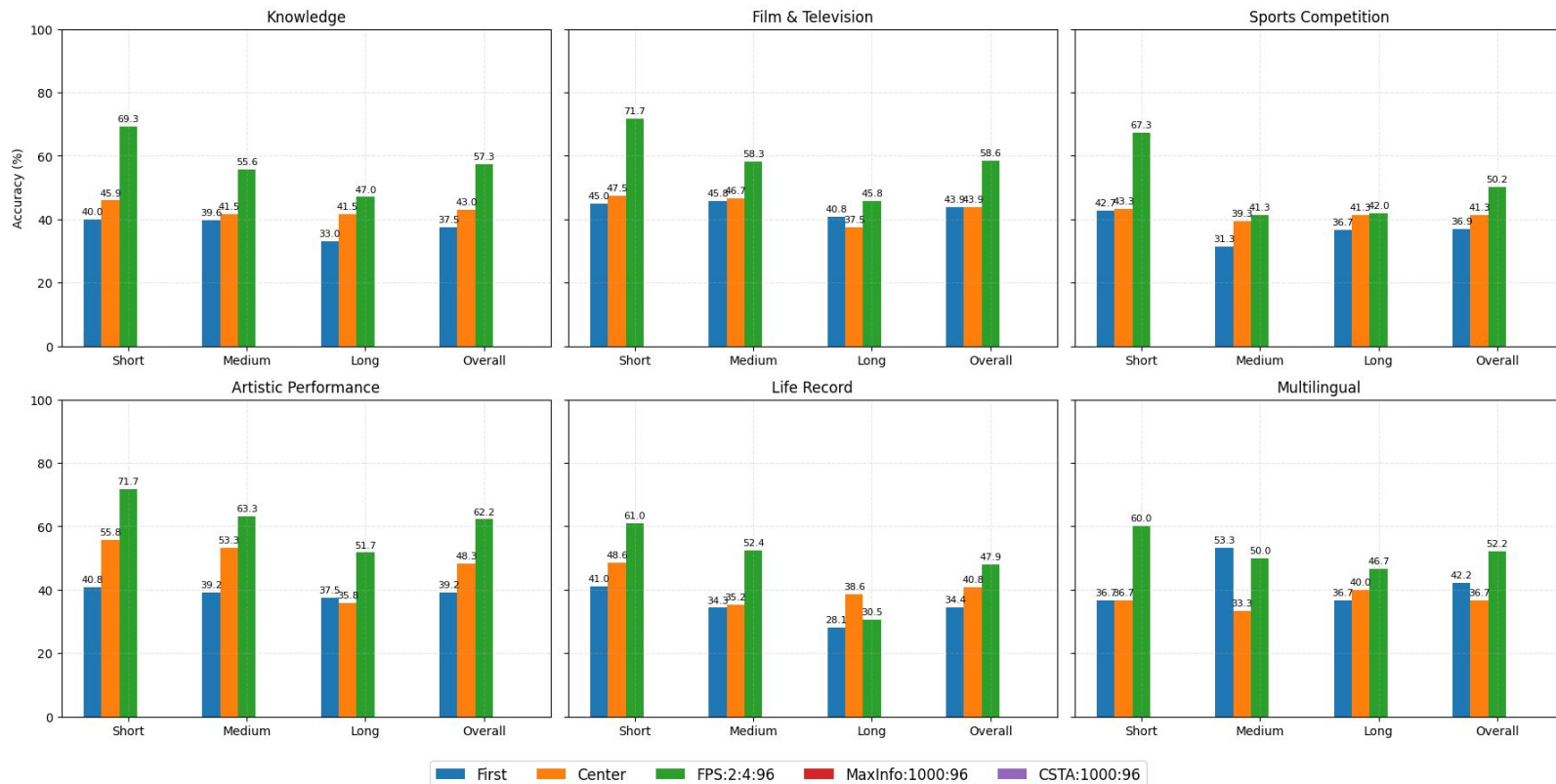
Table 2: VideoMME results over frame sampling techniques: Model: InternVL3

InternVL3-2B



InternVL3-2B

InternVL3 performance on VideoMME for Video Categories and Sampling Techniques



Models Comparison

Video Type	First	Center	FPS	MaxInfo	CSTA
Qwen2.5	41.9%	43.5%	60.4%	57.5%	59.4%
SmolVLM	36.2%	38.4%	39.9%	48.7%	38.5%
InternVL3	37.9%	42.8%	54.6%	%	%
Ovis2	%	%	%	%	%

Table 4: VideoMME results over frame sampling techniques: Model: SmolVLM